WELLSENSE · MARKET AND VIABILITY

WellSense market and viability

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade), with cited benchmark prices checked on 2026-10-25

WellSense market and viability

WellSense is for the people who share an aquifer and have no measurement of it: village water committees, farmer groups, handpump caretakers and the NGOs and agencies that work with them, first in India with a participatory groundwater partner. A commercial level logger without telemetry lists at USD 425 to 750 and an Indian telemetered recorder at INR 47,500, while WellSense's parts cost USD 264.40 at a 30 m depth (USD 403.40 with the lab's FieldNode telemetry core), and it is open, so a community or a school can build, check and repair it. Prices and benchmarks are cited with the date they were checked. Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Participatory groundwater management	Farmer groups and village water committees sharing an aquifer; India first (ACWADAM, Pune, first candidate partner, not yet approached)	CGWA-style digital water level recorders from an Indian maker: INR 34,000 to 38,000 without telemetry and INR 47,500 with telemetry (Acorn Controls listings, checked 2026-10-25)	264.40 parts at 30 m; 403.40 with the FieldNode core and LoRaWAN	Continuous, community-owned record at a fraction of a commercial recorder; open data; repairable locally
WASH programmes and handpump networks	NGOs and utilities in sub-Saharan Africa and South Asia monitoring community water points	Onset HOBO U20L-04 water level logger, 0 to 4 m: USD 425.00 (Instrumart, checked 2026-10-25), no telemetry and a separate barometric logger needed	264.40 (233.40 where no pump shares the casing and no tube is needed)	Telemetry included through FieldNode; early warning when the level nears the pump intake
Network densification for agencies	Groundwater departments and researchers adding wells between official monitoring stations	Solinst Levelogger 6: USD 749.55; Levelogger 6 Junior: USD 535.80 (Fondriest, checked 2026-10-25), no telemetry	264.40 parts (403.40 with telemetry)	More wells for the same budget; open hardware and data formats
Teaching and research	Universities and schools teaching hydrogeology and instrumentation	Same commercial loggers as above	264.40 parts	Inspectable, buildable instrument for teaching; a lab exercise in calibration against a tape

The benchmarks are list prices for one unit from distributors, not quotes; they buy a sealed, calibrated, supported instrument with years of field record, which WellSense does not yet have (PR-004 at risk, PR-005 to verify at TRL 4). WellSense's figures are indicative estimates from `bom/bom.csv` (September 2026, quantity 1), not quotes. The USD 200 value-engineering target is a control target, not a price.

Open route

- **Who builds it.** A partner's field team, a district workshop or a university lab, from the build plan (WLS-BLD-001) with plumbing, electrical and online parts and hand tools; the FieldNode core comes from the lab's shared design.

- **How it spreads.** Through participatory groundwater programmes (the first candidate partner works with farmer groups on shared aquifers), WASH NGOs, the Helpful Engineering volunteer network and hydrogeology courses; build reports and design suggestions come back through the repo's issue templates.
- **What keeps it open.** CERN-OHL-S-2.0 for the hardware and MIT for the software (REUSE.toml), data owned by the hosting community (PR-014), and a design that avoids proprietary parts: any vented 4 to 20 mA transducer of the right size and class will do.
- **What it is not.** It is not a certified instrument for regulatory compliance (WLS-REG-001 G6, G7); a regulated user would need a certified recorder until WellSense has a test record and, if pursued, a certification route.

Sources

- Instrumart. "HOBO U20L Series Water Level Loggers." https://www.instrumart.com/products/40734/hobo-u20l-series-water-level-loggers (U20L-04 at USD 425.00; checked 2026-10-25).
- Fondriest Environmental. "Solinst Levelogger 6 Water Level Loggers." https://fondriest.com/solinst-levelogger-5-water-level-loggers.htm (Levelogger 6 at USD 749.55, Levelogger 6 Junior at USD 535.80; checked 2026-10-25).
- Acorn Controls Private Limited, IndiaMART listings. "Digital Piezometer." https://m.indiamart.com/acorn-controls/digital-piezometer.html (DWLR and piezometer sets INR 34,000 to 47,500; telemetry piezometer DWLR INR 47,500; checked 2026-10-25).
- Central Ground Water Authority. Notification S.O. 3289(E), 24 September 2020, Table 14.1 (see WLS-REG-001 G6).
- WellSense cost: `bom/bom.csv` and WLS-CAL-001 [H2], [H3].